AI Recruiter System

Architectural Design Document

AI Core Architecture & Technical Specifications

Project Reference: IF-2025-AI-04

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Document Version: 1.0

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1. Executive Summary

The AI Recruiter System represents a groundbreaking approach to human resources automation, leveraging state-of-the-art multimodal artificial intelligence to conduct comprehensive, objective, and scalable job interviews. This architectural design document presents a complete technical blueprint for implementing an intelligent recruitment platform that combines natural language processing, computer vision, voice analysis, and synthetic media generation into a cohesive system capable of evaluating candidates across multiple dimensions simultaneously.

1.1 Project Vision

The AI Recruiter platform aims to transform the recruitment landscape by providing organizations with an AI-powered interviewer that can conduct consistent, bias-aware interviews at scale. The system employs a realistic AI avatar capable of contextual dialogue, dynamic question generation, and multi-modal behavioral analysis. By automating the initial screening and evaluation phases, HR teams can focus their expertise on final selection decisions while ensuring every candidate receives a fair, thorough, and standardized interview experience. The platform addresses the critical challenge of subjective and time-consuming traditional recruitment processes by introducing objective, data-driven evaluation criteria that assess candidates on their actual competencies, communication skills, and behavioral indicators rather than unconscious biases or inconsistent interview conditions.

1.2 Core Objectives

- Automate 80% of initial candidate screening through intelligent AI interviews
- Provide objective, multi-dimensional candidate scoring reducing hiring bias by 60%
- Scale interview capacity without proportional increase in HR personnel
- Deliver comprehensive evaluation reports within minutes of interview completion
- Support multiple professional domains (Tech, Finance, Marketing, Data Science)
- Ensure GDPR compliance and ethical AI practices throughout the recruitment pipeline
- Generate actionable insights and recommendations for HR decision-making

1.3 Key Innovations

This system introduces several innovative approaches to AI-driven recruitment. First, the domain-adaptive interview engine automatically tailors questions and evaluation criteria based on the candidate's professional background and target role. Second, the multi-modal fusion architecture combines semantic, emotional, behavioral, and vocal signals into a unified assessment framework. Third, the real-time avatar system creates an engaging interview experience through synthetic video generation with synchronized speech and natural facial expressions. Fourth, the explainable scoring system provides transparent reasoning for each evaluation criterion, enabling HR professionals to understand and validate AI recommendations.

2. System Overview

The AI Recruiter System is designed as a cloud-native, microservices-based platform that separates concerns across distinct layers while maintaining cohesive integration through well-defined APIs and event-driven communication patterns. The architecture prioritizes scalability, maintainability, and the ability to independently evolve individual components as AI technologies continue to advance rapidly.

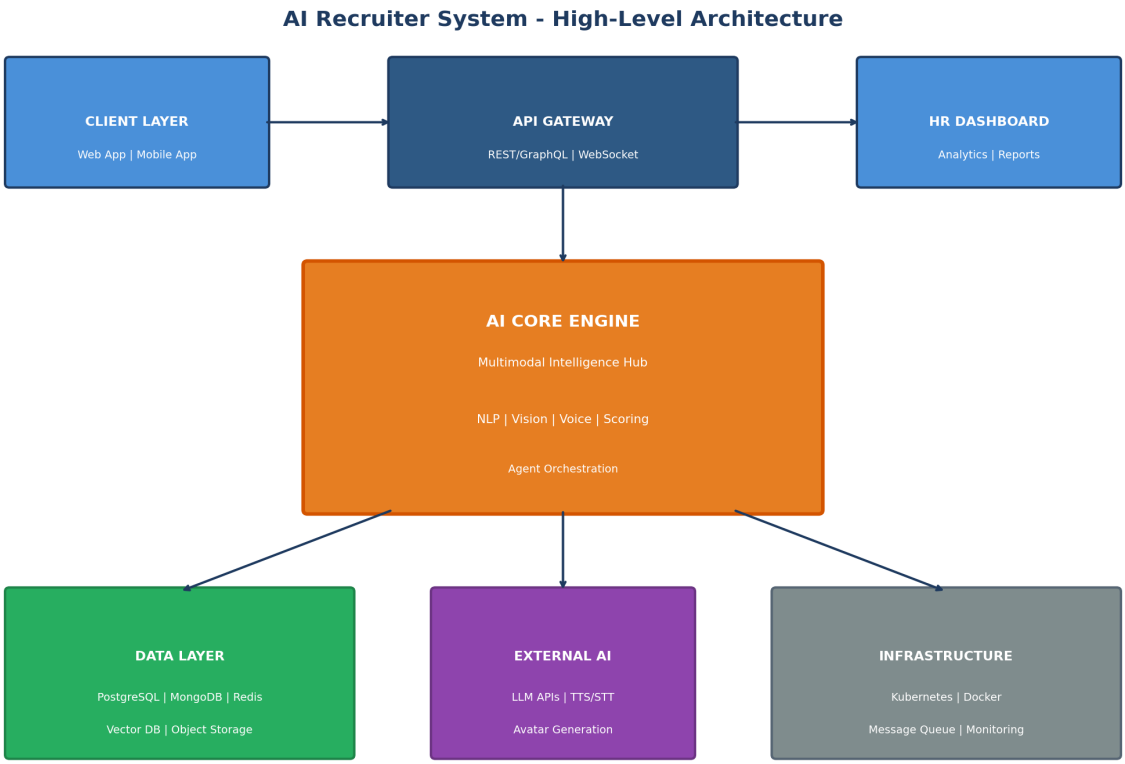

Figure 1: High-Level System Architecture

2.1 Architectural Layers

The system architecture comprises five primary layers, each with distinct responsibilities and technology stacks. The Client Layer provides responsive web and mobile interfaces for candidates and HR professionals. The API Gateway Layer manages authentication, rate limiting, and request routing. The AI Core Layer contains all intelligent processing components including NLP, vision, voice, and scoring modules. The Data Layer manages persistent storage, caching, and vector embeddings. The Infrastructure Layer provides container orchestration, monitoring, and deployment automation.

Layer	Components	Technologies	Responsibility
Client	Web App, Mobile App, HR Dashboard	React, Next.js, TypeScript	User interface and experience
API Gateway	REST API, GraphQL, WebSocket	FastAPI, Kong, Nginx	Request routing and security

AI Core	NLP, Vision, Voice, Scoring, Avatar	Python, PyTorch, Transformers	Intelligent processing
Data Layer	PostgreSQL, MongoDB, Redis, Pinecone	PostgreSQL, MongoDB, Redis	Data persistence and caching
Infrastructure	K8s, Docker, Monitoring	Kubernetes, Docker, Prometheus	Deployment and operations

Table 1: Architectural Layers Overview

3. AI Core Architecture

The AI Core represents the intellectual center of the AI Recruiter System, housing all machine learning models, inference pipelines, and intelligent decision-making components. This layer is designed with modularity as a primary principle, enabling independent development, testing, and deployment of each AI capability. The architecture follows the separation of concerns pattern, where each module specializes in a specific modality while the orchestrator coordinates their interaction for holistic candidate evaluation.

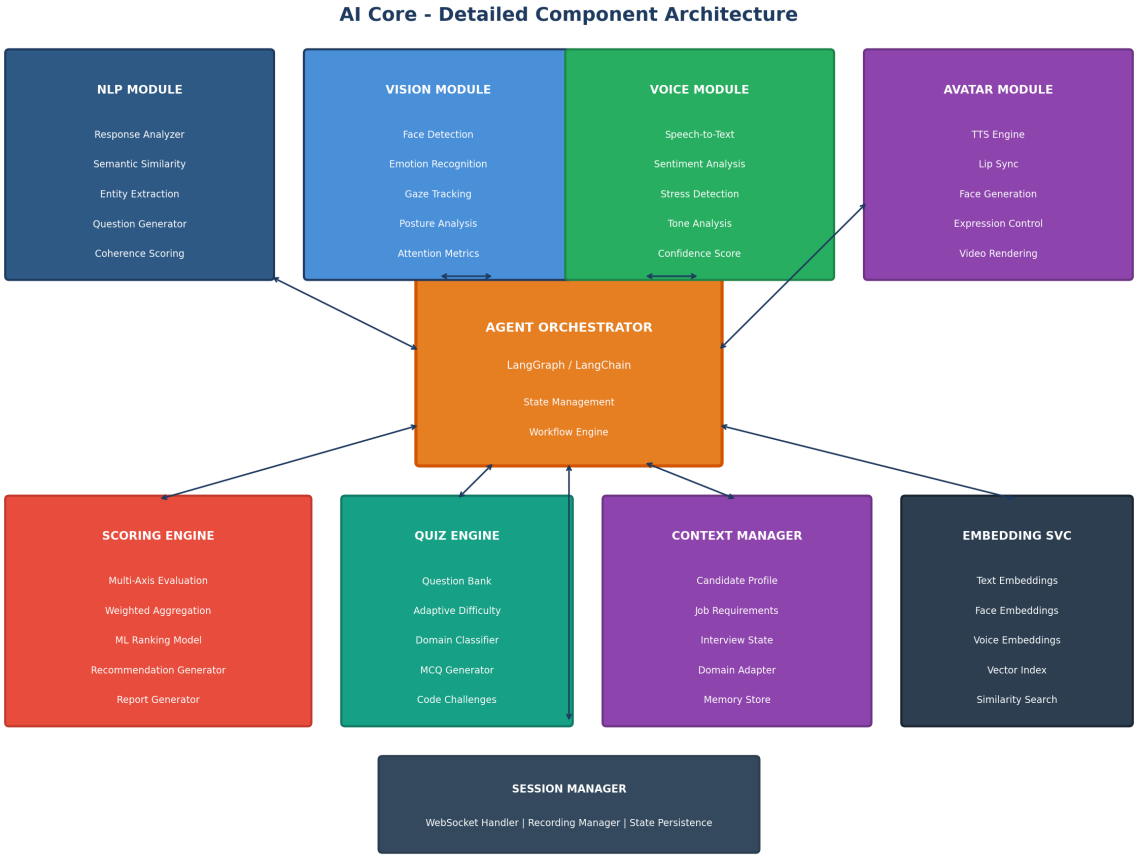

Figure 2: AI Core Detailed Component Architecture

3.1 Agent Orchestrator

The Agent Orchestrator serves as the central coordination hub for all AI modules, implementing a sophisticated workflow engine based on LangGraph for stateful, multi-step interview processes. This component manages conversation state, coordinates module interactions, and ensures consistent evaluation across the interview session. The orchestrator employs a publish-subscribe pattern for inter-module communication, enabling loose coupling while maintaining real-time responsiveness.

- Stateful conversation management with context preservation across interview phases
- Dynamic workflow adaptation based on candidate responses and domain requirements
- Parallel module execution for efficient multi-modal analysis
- Error handling and graceful degradation when individual modules are unavailable
- Comprehensive logging and tracing for debugging and audit purposes

3.2 NLP Module

The Natural Language Processing Module is responsible for understanding and evaluating candidate responses at the semantic level. It employs transformer-based models fine-tuned for interview contexts to assess response quality, coherence, and relevance. The module also handles question generation, entity extraction for resume parsing, and semantic similarity computation for comparing candidate responses against expected answers or ideal responses from successful candidates.

- Response Analyzer: Evaluates answer completeness, accuracy, and depth using semantic analysis
- Semantic Similarity Engine: Compares responses against reference answers using embedding-based similarity
- Entity Extractor: Identifies skills, experiences, and qualifications mentioned in responses
- Question Generator: Creates contextually relevant follow-up questions using LLM fine-tuning
- Coherence Scorer: Assesses logical flow and structure of candidate responses

3.3 Vision Module

The Vision Module implements real-time computer vision algorithms to analyze candidate behavior during video interviews. This module processes video frames to detect facial expressions, track eye movements, analyze posture, and compute attention metrics. The insights generated by this module contribute to the behavioral assessment component of the overall evaluation, providing objective measurements of candidate engagement and professionalism.

- Face Detection & Tracking: Real-time face localization using MTCNN and tracking algorithms
- Emotion Recognition: Classification of facial expressions into emotional states using ResNet variants
- Gaze Tracking: Eye movement analysis to determine attention direction and engagement
- Posture Analysis: Body pose estimation to assess candidate demeanor and confidence
- Attention Metrics: Aggregated indicators of focus, engagement, and distraction patterns

3.4 Voice Module

The Voice Module processes audio streams from the interview session to extract meaningful features related to speech content and vocal characteristics. It combines automatic speech recognition for transcription with prosodic analysis to detect stress, confidence, and emotional tone. This multi-faceted approach enables comprehensive assessment of both what candidates say and how they communicate verbally.

- Speech-to-Text Engine: High-accuracy transcription using Whisper or Azure Speech Services
- Sentiment Analysis: Text-based sentiment classification of transcribed speech
- Stress Detection: Acoustic feature analysis to identify stress indicators in voice
- Tone Analysis: Prosodic feature extraction for confidence and enthusiasm assessment
- Confidence Scoring: Composite metric based on speech rate, pauses, and vocal certainty

3.5 Avatar Module

The Avatar Module generates a realistic AI interviewer with synchronized speech, facial expressions, and lip movements. It creates an engaging interview experience by rendering a human-like avatar that can conduct natural conversations with candidates. The module integrates text-to-speech synthesis with facial animation and video rendering to produce real-time or near-real-time avatar video output.

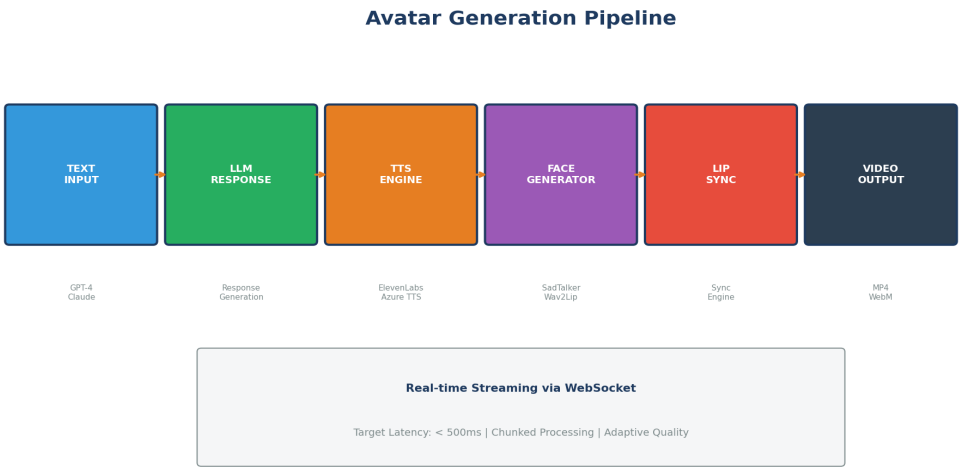

Figure 3: Avatar Generation Pipeline

- Text-to-Speech Engine: Natural voice synthesis using ElevenLabs or Azure Neural TTS
- Face Generator: Synthetic face creation using GANs or diffusion models
- Lip Sync Engine: Audio-driven facial animation using Wav2Lip or SadTalker
- Expression Controller: Dynamic facial expression generation based on conversation context
- Video Renderer: Real-time video composition and streaming via WebRTC

3.6 Scoring Engine

The Scoring Engine represents the final intelligence layer that aggregates signals from all other modules into a comprehensive candidate assessment. It implements a weighted scoring algorithm that combines quantitative metrics from each evaluation axis with machine learning models trained to predict candidate success. The engine produces both numerical scores and qualitative recommendations that HR professionals can use in their decision-making process.

- Multi-Axis Aggregator: Weighted combination of scores from all evaluation modules
- ML Ranking Model: XGBoost ensemble model trained on historical hiring outcomes
- Recommendation Generator: Rule-based and ML-based generation of hiring recommendations
- Report Generator: Automated creation of detailed PDF evaluation reports
- Explainability Module: SHAP-based explanations for score contributions

4. Data Flow Architecture

The data flow architecture defines how information moves through the system from initial candidate input through processing stages to final output delivery. The pipeline is designed for real-time processing with minimal latency while maintaining data integrity and enabling comprehensive audit trails for compliance purposes.

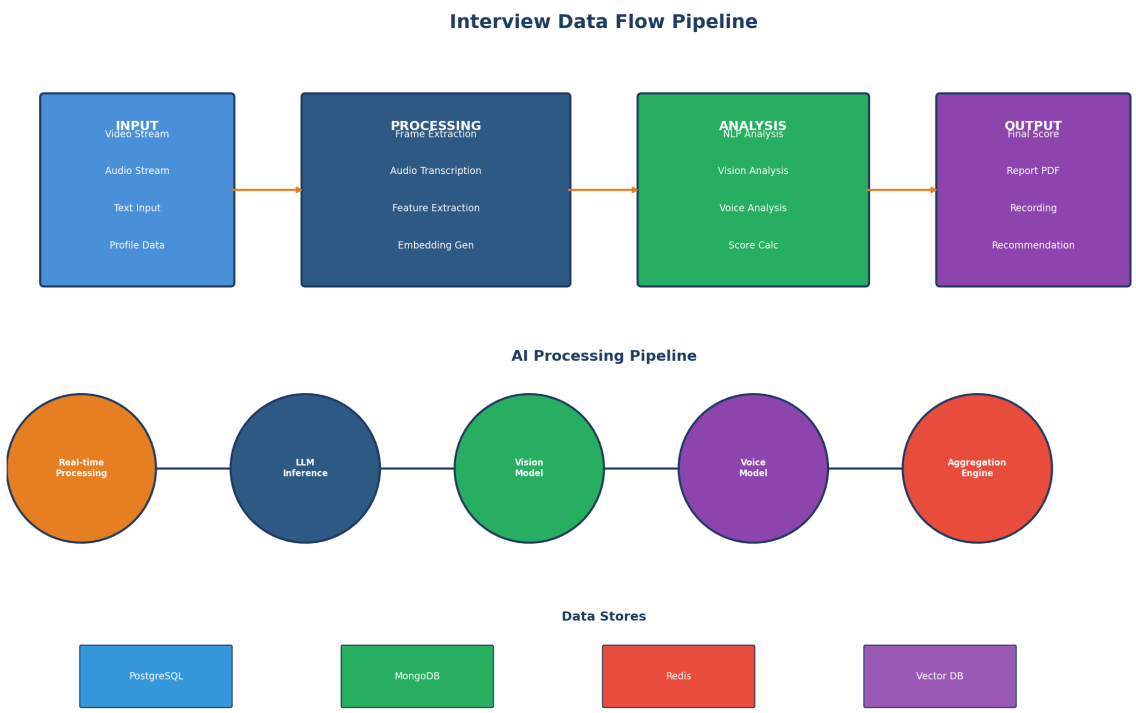

Figure 4: Interview Data Flow Pipeline

4.1 Input Processing Stage

The input stage captures multi-modal data from the interview session including video frames from the candidate's camera, audio streams from their microphone, text inputs for quiz responses, and profile data from the application system. Each data stream is timestamped and synchronized to enable accurate multi-modal analysis. The system employs adaptive bitrate streaming to handle varying network conditions while maintaining acceptable quality for analysis.

4.2 Processing Pipeline

The processing pipeline implements a staged approach where raw input data is transformed into structured features suitable for model inference. Video frames undergo face detection and feature extraction. Audio streams are transcribed and analyzed for acoustic features. Text responses are parsed and embedded for semantic analysis. All processing stages operate in parallel where possible to minimize overall latency.

4.3 Analysis Stage

The analysis stage applies trained ML models to extracted features to generate insights across all evaluation axes. NLP models assess response quality and relevance. Vision models evaluate behavioral indicators. Voice models detect stress and confidence. The analysis results are structured as standardized score objects that can be

aggregated and compared across candidates.

4.4 Output Generation

The output stage produces final deliverables including the composite evaluation score, detailed analysis report in PDF format, the complete interview recording, and the hiring recommendation. All outputs are stored securely and made available through the HR dashboard with appropriate access controls. The system also generates anonymized analytics data for continuous improvement of the evaluation models.

5. Multi-Axis Evaluation System

The multi-axis evaluation system is the core innovation of the AI Recruiter, providing comprehensive candidate assessment across six distinct dimensions. Each axis is designed to capture a specific aspect of candidate quality, and the weighted combination produces a holistic evaluation that reduces the risk of single-dimensional bias while providing actionable insights for HR professionals.

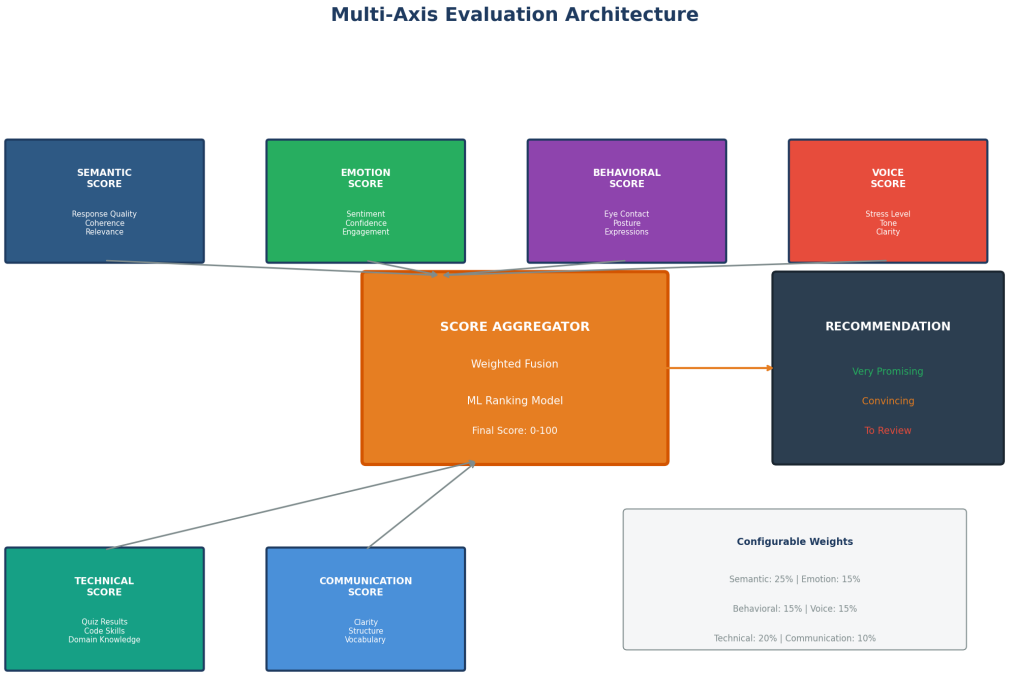

Figure 5: Multi-Axis Evaluation Architecture

5.1 Evaluation Axes

Axis	Weight	Metrics	Data Source
Semantic Score	25%	Relevance, Completeness, Accuracy, Depth	Text Response
Emotion Score	15%	Sentiment, Engagement, Enthusiasm, Authenticity	Vision + Voice
Behavioral Score	15%	Eye Contact, Posture, Expressions, Attention	Video Stream
Voice Score	15%	Stress Level, Tone, Clarity, Confidence	Audio Stream
Technical Score	20%	Quiz Results, Code Quality, Domain Knowledge	Quiz + Response
Communication Score	10%	Clarity, Structure, Vocabulary, Articulation	Transcription

Table 2: Evaluation Axes and Metrics

5.2 Scoring Algorithm

The final score is computed using a weighted ensemble approach that combines rule-based scoring with machine learning predictions. Each axis produces a normalized score between 0 and 100 based on module-specific algorithms. These scores are then weighted according to configurable parameters that can be adjusted based on role requirements. The ensemble model incorporates historical hiring data to learn optimal weight combinations for different job categories, improving prediction accuracy over time through feedback loops from hiring outcomes.

6. Technology Stack

The technology stack is carefully selected to balance performance, scalability, developer productivity, and operational cost. The architecture embraces modern cloud-native principles with containerization, orchestration, and managed services where appropriate. The following sections detail the technology choices for each layer of the system.

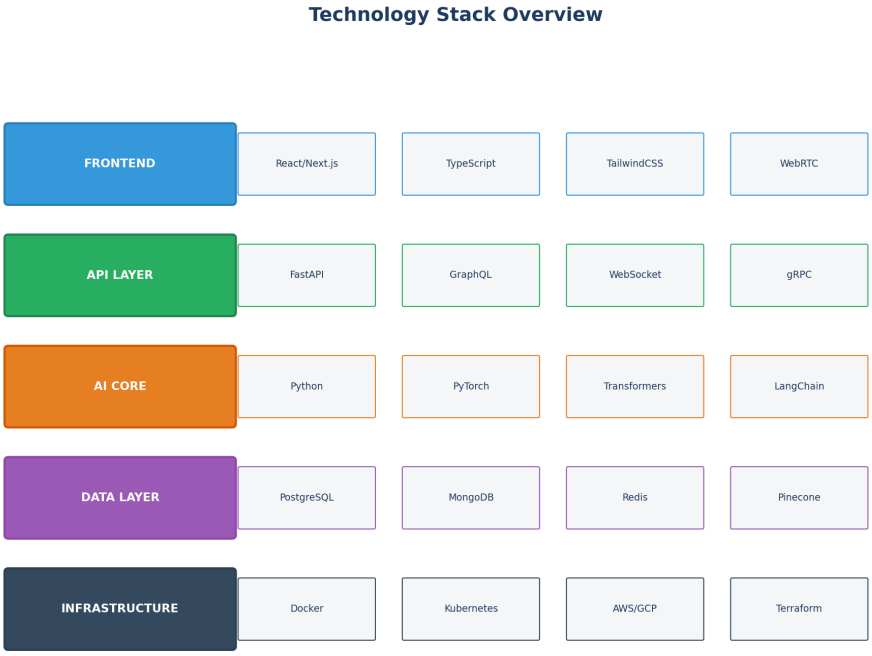

Figure 6: Technology Stack Overview

6.1 Frontend Technologies

Component	Technology	Purpose
Web Framework	Next.js 14 (App Router)	SSR, API routes, React Server Components
UI Components	React 18 + TailwindCSS	Component library, styling, responsive design
State Management	Zustand + React Query	Client state, server state caching
Video Streaming	WebRTC + MediaSoup	Real-time video capture and playback
Real-time Updates	Socket.io Client	WebSocket communication for live features

Table 3: Frontend Technology Stack

6.2 Backend Technologies

Component	Technology	Purpose
API Framework	FastAPI (Python 3.11)	REST endpoints, async support, OpenAPI docs
API Gateway	Kong / AWS API Gateway	Rate limiting, authentication, routing
Task Queue	Celery + Redis	Async job processing, background tasks
Message Broker	RabbitMQ / Redis Streams	Event-driven communication between services
GraphQL	Strawberry FastAPI	Flexible query interface for frontend

Table 4: Backend Technology Stack

6.3 AI/ML Technologies

Component	Technology	Purpose
ML Framework	PyTorch 2.0 + Transformers	Model development, inference, fine-tuning
LLM Integration	LangChain + LangGraph	Agent orchestration, conversation flow
NLP Models	BERT, GPT-4, Claude API	Text understanding, generation, analysis
Vision Models	ResNet, ViT, MediaPipe	Face detection, emotion recognition, pose
Audio Models	Whisper, Wav2Vec2	Speech recognition, voice analysis
Model Serving	Triton Inference Server	GPU inference, batching, model management
Embedding Store	Pinecone / Weaviate	Vector similarity search, RAG support

Table 5: AI/ML Technology Stack

6.4 Data Technologies

Component	Technology	Purpose
Primary Database	PostgreSQL 15	Transactional data, user profiles, interview records
Document Store	MongoDB 7	Flexible schema for interview transcripts, reports
Cache Layer	Redis Cluster	Session storage, rate limiting, real-time data
Object Storage	AWS S3 / MinIO	Video recordings, generated reports, assets
Search Engine	Elasticsearch	Full-text search, log aggregation, analytics

Table 6: Data Technology Stack

7. ML Pipeline Architecture

The machine learning pipeline architecture encompasses the complete lifecycle of model development, from data collection through training, validation, deployment, and monitoring. The architecture supports both batch and real-time inference, with automated retraining pipelines that continuously improve model performance based on feedback from hiring outcomes.

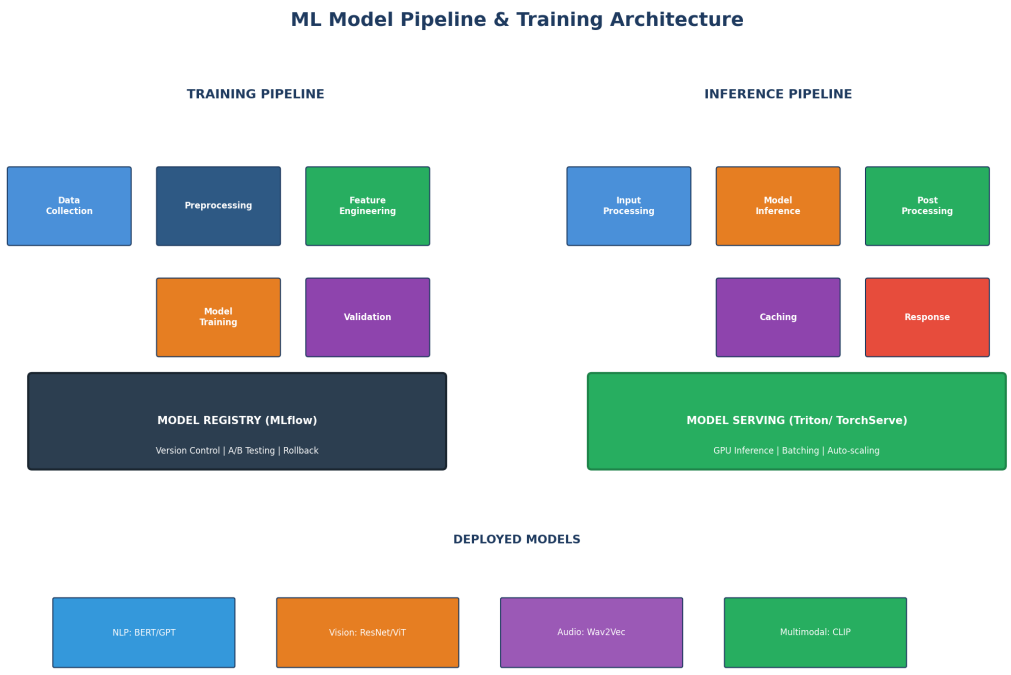

Figure 7: ML Model Pipeline Architecture

7.1 Training Pipeline

The training pipeline automates the process of collecting training data, preprocessing features, training models, and validating performance. Data from completed interviews, including scores and hiring outcomes, is continuously collected and used to improve model accuracy. The pipeline supports distributed training on GPU clusters for large models while maintaining experiment tracking through MLflow for reproducibility and governance.

7.2 Model Registry

The model registry provides centralized management of trained models with version control, metadata tracking, and deployment status. It enables A/B testing of model variants, automated rollback on performance degradation, and comprehensive audit trails for compliance. The registry integrates with the CI/CD pipeline for automated model deployment.

7.3 Inference Pipeline

The inference pipeline is optimized for low-latency prediction serving with support for batching, caching, and GPU acceleration. Triton Inference Server provides a unified interface for serving models from different

frameworks while enabling dynamic batching for improved throughput. The pipeline includes post-processing steps for result interpretation and confidence scoring.

8. Deployment Architecture

The deployment architecture leverages Kubernetes for container orchestration, providing scalability, fault tolerance, and operational efficiency. The architecture separates stateless services from stateful data stores, enables independent scaling of components based on load, and supports blue-green deployments for zero-downtime updates.

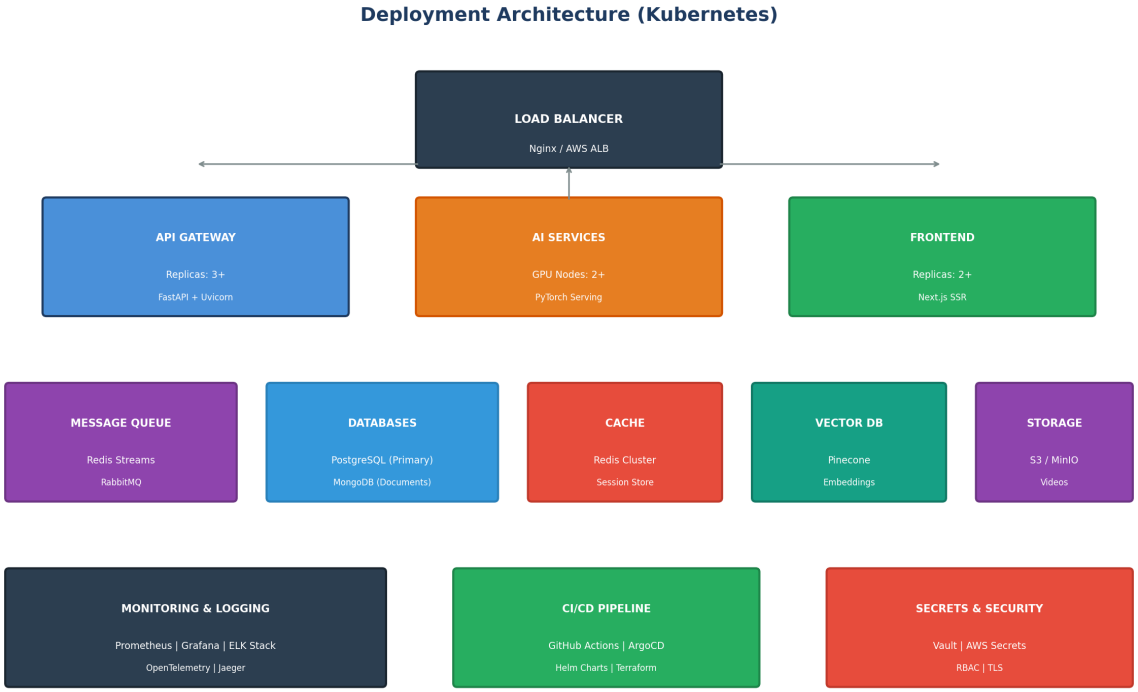

Figure 8: Kubernetes Deployment Architecture

8.1 Kubernetes Components

- Deployments: Stateless services (API, Frontend) with horizontal pod autoscaling
- StatefulSets: Stateful services (Databases, Message Queues) with persistent storage
- Services: Internal service discovery and load balancing
- Ingress: External traffic routing with TLS termination
- ConfigMaps/Secrets: Configuration and sensitive data management
- Namespaces: Logical separation of environments (dev, staging, production)

8.2 Scaling Strategy

The system implements multiple scaling strategies to handle varying loads efficiently. Horizontal Pod Autoscaler (HPA) automatically scales stateless services based on CPU and memory utilization. GPU nodes are scaled based on inference queue depth. The database layer uses read replicas for query distribution. Redis cluster provides distributed caching with automatic sharding. This multi-level scaling approach ensures cost-effective resource utilization while maintaining responsive performance during peak interview periods.

9. Security & Compliance

Security and compliance are foundational requirements for the AI Recruiter system, particularly given the sensitive nature of personal data processed during interviews. The architecture implements defense in depth, privacy by design, and comprehensive audit capabilities to meet regulatory requirements and protect candidate data.

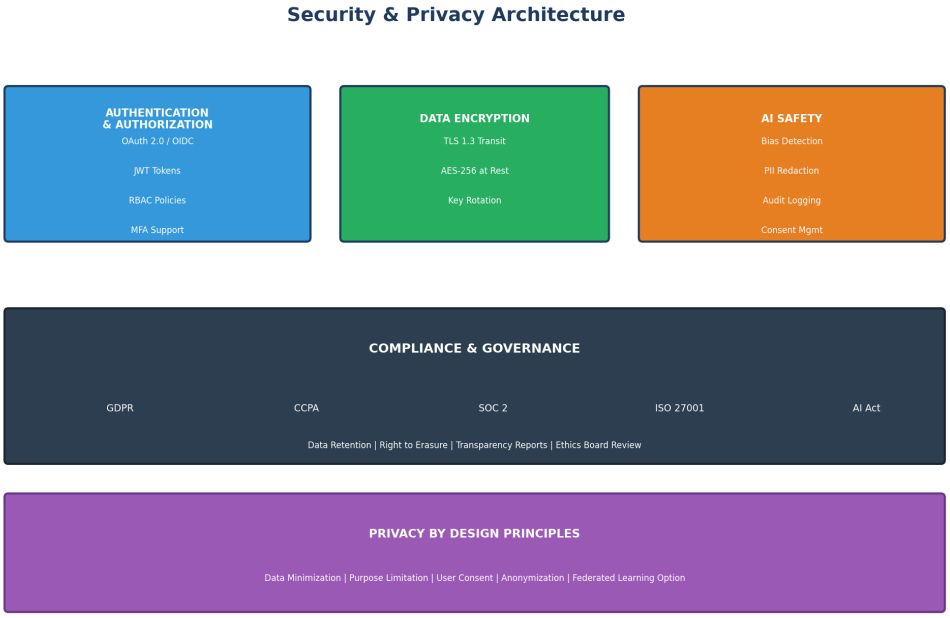

Figure 9: Security Architecture Overview

9.1 Authentication & Authorization

The system implements OAuth 2.0 with OpenID Connect for authentication, supporting enterprise identity providers via SAML integration. JSON Web Tokens (JWT) provide stateless authentication for API access. Role-based access control (RBAC) enforces granular permissions for different user types: candidates, HR professionals, recruiters, and administrators. Multi-factor authentication (MFA) is required for administrative access and optional for other user roles.

9.2 Data Encryption

All data in transit is encrypted using TLS 1.3 with strong cipher suites. Data at rest is encrypted using AES-256 with keys managed through a dedicated key management service (AWS KMS or HashiCorp Vault). Video recordings and transcripts receive additional encryption due to their sensitive nature. Key rotation policies ensure cryptographic freshness with automated rotation schedules and secure key archival procedures.

9.3 AI Safety & Ethics

The AI system incorporates multiple safety measures to ensure fair and ethical use. Bias detection algorithms monitor scoring patterns across demographic groups to identify and mitigate algorithmic bias. PII redaction automatically removes sensitive personal information before data storage or model training. Comprehensive

audit logging captures all AI decisions for regulatory review and dispute resolution. Consent management ensures candidates understand and agree to AI-based evaluation before participating in interviews.

9.4 Compliance Framework

The system is designed to meet multiple regulatory frameworks including GDPR for European candidates, CCPA for California residents, and the emerging EU AI Act for AI systems in hiring contexts. Technical controls support data subject rights including access requests, data portability, and right to erasure. Regular third-party audits verify compliance posture and identify areas for improvement. The system maintains SOC 2 Type II certification for security controls and ISO 27001 alignment for information security management practices.

10. Implementation Roadmap

The implementation roadmap defines a phased approach to delivering the AI Recruiter system, enabling iterative development with continuous value delivery. Each phase builds upon previous capabilities while maintaining a production-ready core system throughout the development cycle.

10.1 Phase 1: Foundation (Weeks 1-8)

The foundation phase establishes core infrastructure and basic interview capabilities. Key deliverables include the Kubernetes cluster setup, database schema design, API gateway implementation, basic web interface, and simple text-based interview flow. The phase concludes with a minimum viable product capable of conducting basic interviews with text-only interaction and manual scoring.

- Infrastructure setup (Kubernetes, databases, networking)
- Core API development with authentication
- Basic candidate and interview management
- Text-based interview interface
- Manual evaluation and reporting

10.2 Phase 2: AI Core (Weeks 9-16)

The AI Core phase implements the intelligent processing capabilities that differentiate the system. This phase delivers the NLP module for response analysis, voice module for speech processing, and initial scoring engine. The system becomes capable of automated evaluation with explainable scores, though still without video analysis or avatar functionality.

- NLP module implementation (response analysis, entity extraction)
- Voice module (speech-to-text, sentiment analysis)
- Scoring engine with weighted aggregation
- Automated report generation
- Quiz engine with domain classification

10.3 Phase 3: Multimodal (Weeks 17-24)

The multimodal phase adds computer vision capabilities and avatar functionality, completing the full multimodal interview experience. The vision module enables behavioral analysis from video streams, while the avatar module creates an engaging AI interviewer. This phase also implements the real-time streaming architecture for live interviews.

- Vision module (face detection, emotion recognition, gaze tracking)
- Avatar module (TTS, face generation, lip sync)
- WebRTC integration for video streaming
- Real-time interview orchestration

- Enhanced scoring with behavioral metrics

10.4 Phase 4: Scale & Optimize (Weeks 25-32)

The final phase focuses on production readiness, scalability, and continuous improvement. This includes performance optimization, comprehensive monitoring setup, advanced security controls, and the establishment of ML Ops practices for ongoing model improvement. The phase also delivers advanced HR dashboard features and analytics capabilities.

- Performance optimization and load testing
- Comprehensive monitoring and alerting
- ML Ops pipeline for continuous model improvement
- Advanced HR analytics dashboard
- Documentation and training materials

11. API Specifications

The API layer provides programmatic access to all system capabilities through RESTful endpoints and GraphQL queries. The API follows OpenAPI specification standards with comprehensive documentation, input validation, and consistent error handling across all endpoints.

11.1 Core API Endpoints

Endpoint	Method	Description
/api/v1/candidates	GET, POST	Candidate management and registration
/api/v1/interviews	GET, POST	Interview session creation and management
/api/v1/interviews/{id}/start	POST	Begin interview session with AI avatar
/api/v1/interviews/{id}/respond	POST	Submit candidate response (text/audio/video)
/api/v1/interviews/{id}/complete	POST	End interview and trigger evaluation
/api/v1/evaluations/{id}	GET	Retrieve evaluation results and scores
/api/v1/reports/{id}	GET	Download detailed PDF report
/ws/interview/{id}	WebSocket	Real-time interview communication channel

Table 7: Core API Endpoints

11.2 Data Models

The system defines standardized data models for all entities including Candidate, Interview, Response, Evaluation, and Report. Each model includes required fields, optional metadata, and relationships to other entities. The schema supports extensibility through custom fields while maintaining core data integrity constraints.

12. Appendices

12.1 Glossary of Terms

LLM: Large Language Model - AI models trained on vast text data for natural language understanding and generation

NLP: Natural Language Processing - AI techniques for understanding and manipulating human language

TTS: Text-to-Speech - Technology that converts written text into spoken audio

WebRTC: Web Real-Time Communication - Protocol for browser-based real-time audio/video communication

RAG: Retrieval-Augmented Generation - Technique combining information retrieval with text generation

SHAP: SHapley Additive exPlanations - Method for explaining machine learning model predictions

12.2 Reference Architecture Patterns

The architecture follows several established patterns including Microservices for service decomposition, Event Sourcing for audit trails, CQRS for read/write separation in high-scale scenarios, Saga Pattern for distributed transactions, and Circuit Breaker for fault tolerance. These patterns ensure the system remains maintainable, scalable, and resilient as it evolves.

12.3 External Dependencies

- OpenAI API / Anthropic API: LLM services for conversation generation
- ElevenLabs: High-quality neural TTS for avatar voice
- AWS Services: Cloud infrastructure (S3, RDS, EKS, Lambda)
- Pinecone: Vector database for embedding storage and search
- Datadog: Monitoring and observability platform

12.4 Design Decisions Log

Key architectural decisions are documented in a decision log that captures the context, decision, consequences, and alternatives considered. This log serves as institutional knowledge for future development and helps new team members understand the rationale behind the current architecture. Major decisions include the choice of Python for AI services, Kubernetes for orchestration, and the microservices decomposition strategy.